

习题 =

2. 解: 令 $f(x) = 1$, 则 $\frac{4}{3}f(\frac{1}{3}) + \frac{1}{4}f(1) = 1 = \int_0^1 f(x)dx$

令 $f(x) = x$, 则 $\int_0^1 f(x)dx = \frac{1}{2}$

$\frac{3}{4}f(\frac{1}{3}) + \frac{1}{4}f(1) = \frac{1}{2} = \int_0^1 f(x)dx$

令 $f(x) = x^2$, 则 $\int_0^1 f(x)dx = \frac{1}{3}$

$\frac{3}{4}f(\frac{1}{3}) + \frac{1}{4}f(1) = \frac{3}{4} \times \frac{1}{9} + \frac{1}{4} = \frac{1}{3} = \int_0^1 f(x)dx$

令 $f(x) = x^3$, 则 $\int_0^1 f(x)dx = \frac{1}{4}$

$\frac{3}{4}f(\frac{1}{3}) + \frac{1}{4}f(1) = \frac{5}{12} \neq \int_0^1 f(x)dx$

$\therefore$ 此求积公式的代数精度为 2.

8. 解: 辛甫生法.

$$S = \frac{1.5-1.1}{6} [e^{1.1} + 4e^{1.3} + e^{1.5}] = 1.47754$$

复化梯形法

$$\int_{1.1}^{1.5} e^x dx \approx \frac{1.5-1.1}{2} (e^{1.1} + e^{1.3}) + \frac{1.5-1.3}{2} (e^{1.3} + e^{1.5}) = 1.48295$$

10. 解: 复化梯形法 $|R| \leq \frac{1}{2} (\frac{1}{n})^2 e \leq \frac{1}{2} \times 10^{-5}$, 得 $n \geq \frac{212.05}{\sqrt{e}}$, 则划分 213 等分

复化辛甫生法 $|R| \leq \frac{1}{2880} (\frac{1}{n})^4 \leq \frac{1}{2e} \times 10^{-5}$, 得 $n \geq 3.7$, 则划分 8 等分

16. 解: 已知三点高斯公式 $\int_{-1}^1 f(x) dx \approx \frac{5}{9} f(-\sqrt{\frac{3}{5}}) + \frac{8}{9} f(0) + \frac{5}{9} f(\sqrt{\frac{3}{5}})$

对于区间 $[1, 3]$, 作变换 $x = t + 2$

$$\int_1^3 f(x) dx \approx \frac{5}{9} f(-\sqrt{\frac{3}{5}} + 2) + \frac{8}{9} f(2) + \frac{5}{9} f(2 + \sqrt{\frac{3}{5}})$$

25. 解: 已知 $x_0 = 1.0$, $x_1 = 1.1$, $x_2 = 1.2$, $h = 0.1$

$$f'(1.0) \approx \frac{1}{2h} [-3f(1.0) + 4f(1.1) - f(1.2)] = -0.2470$$

$$f'(1.1) \approx \frac{1}{2h} [-f(1.0) + f(1.2)] = -0.2170$$

$$f'(1.2) \approx \frac{1}{2h} [f(1.0) - 4f(1.1) + 3f(1.2)] = -0.1870$$

$$\therefore f(x) = \frac{1}{(1+x)^2}$$

$$\therefore f^{(3)}(x) = -\frac{24}{(1+x)^5}$$

由余项表达式计算误差

$$R(1.0) = \frac{f^{(3)}(\xi)}{3!} h^2 \approx \frac{-24 \times 0.1^2}{3! (1+1.0)^5} \approx -0.0025$$

$$R(1.1) = -\frac{f^{(3)}(\xi)}{3!} h^2 \approx \frac{24 \times 0.1^2}{3! (1+1.0)^5} \approx 0.00125$$

$$R(1.2) = \frac{f^{(3)}(\xi)}{3!} h^2 \approx \frac{-24 \times 0.1^2}{3! (1+1.1)^5} \approx \cancel{-0.001967} -0.001961$$

习题三

解: $y_{n+1} = y_n + hf(x_n, y_n) = y_n + 0.1 \left(\frac{1}{1+x_n} - 2y_n \right)$

计算结果如下:

x_n	y_n	$y(x_n)$	x_n	y_n	$y(x_n)$
0.1	0.10000	0.09901	0.6	0.46557	0.44117
0.2	0.19701	0.19231	0.7	0.49575	0.46980
0.3	0.28540	0.27523	0.8	0.51371	0.48781
0.4	0.360854	0.34483	0.9	0.52190	0.49724
0.5	0.42102	0.40000	1.0	0.52268	0.50000

1. 解:
$$\begin{cases} y_p = y_n + hf(x_n, y_n) \\ y_c = y_n + hf(x_{n+1}, y_p) \\ y_{n+1} = \frac{1}{2}(y_p + y_c) \end{cases}$$

x_n	y_n	$y(x_n)$	x_n	y_n	$y(x_n)$
0.1	0.09851	0.09901	0.6	0.43885	0.44118
0.2	0.19124	0.19231	0.7	0.46716	0.46780
0.3	0.27373	0.27523	0.8	0.48556	0.48781
0.4	0.34293	0.34483	0.9	0.49516	0.49724
0.5	0.39782	0.40000	1.0	0.49813	0.50000

1. 解: 由题意得 k_1, k_2, k_3, k_4 的具体形式是

$$k_1 = 8 - y_n$$

$$k_2 = 8 - (y_n + \frac{h}{2}k_1)$$

$$k_3 = 8 - (y_n + \frac{h}{2}k_2)$$

$$k_4 = 8 - (y_n + \frac{h}{2}k_3)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 1.2016 + 0.5494y_n$$

当 $x_1 = 0.2$ 时, $y_1 = 2.3004$

$x_2 = 0.4$ 时, $y_2 = 2.4654$

1.2 解: ① 欧拉法, $h=0.1$

x_n	y_n				
0.0	1.0000	0.9	1.2875	1.9	2.0351
0.1	1.0000	1.0	1.3488	2.0	2.1216
0.2	1.0100	1.1	1.4139		
0.3	1.0290	1.2	1.4825		
0.4	1.0561	1.3	1.5542		
0.5	1.0905	1.4	1.6288		
0.6	1.1315	1.5	1.7059		
0.7	1.1784	1.6	1.7868		
0.8	1.2306	1.8	1.9501		

改进的欧拉方法:

x_n	y_n	x_n	y_n
0.0	1.0000	1.2	1.5040
0.2	1.0200	1.4	1.6493
0.4	1.0724	1.6	1.8044
0.6	1.1514	1.8	1.9676
0.8	1.2521	2.0	2.1374
1.0	1.3707		

四阶龙格-库塔方法:

x_n	y_n	
0.0	1.0000	7.85.1
0.4	1.0704	8.2.2.1
0.8	1.2444	1.2.1.1
1.2	1.5013	2.8.4.1
1.6	1.8020	2.9.2.1
2.0	2.1354	8.5.1.1
		4.7.1.1

习题四

5. 解: (1) 证明: 由题意得 $\varphi(x) = 4 + \frac{2}{3} \cos x$

$$\varphi'(x) = -\frac{2}{3} \sin x, \text{ 在 } (-\infty, +\infty) \text{ 上恒有 } |\varphi'(x)| \leq \frac{2}{3} < 1$$

由压缩映像原理定理1可证 $\varphi(x)$ 对于任意初值 x_0 均收敛

(2) 证明: $\therefore \varphi'(x) = -\frac{2}{3} \sin x$

由方程 $12 - 3x + 2 \cos x = 0$, 易知 $x = k\pi, k \in \mathbb{Z}$ 不为方程解

$\therefore \varphi'(x^*) \neq 0$, 则 $x_{k+1} = 4 + \frac{2}{3} \cos x_k$ 为线性收敛

k	0	1	2	3
x_k	0.4	4.66665	4.66446	4.66446

近似根为 4.66446

7. 解: (1) 由题意得 迭代公式为 $\varphi(x) = 1 + \frac{1}{x^2}$

$$\varphi'(x) = -\frac{2}{x^3} \therefore \text{当 } x \in [1.3, 1.6] \text{ 时, } |\varphi'(x)| = \frac{2}{x^3} \leq \frac{2}{1.3^3} = \frac{2000}{2197} < 1,$$

$\therefore$ 迭代公式在区间 $[1.3, 1.6]$ 上收敛

$$\text{且 } 1.3 < 1 + \frac{1}{1.6^2} \leq \varphi(x) \leq 1 + \frac{1}{1.3^2} < 1.6$$

(2) 由题意得 迭代公式为 $\varphi(x) = \sqrt[3]{1+x^2}$

$$\varphi'(x) = \frac{1}{3\sqrt[3]{(1+x^2)^2}}, \text{ 当 } x \in [1.3, 1.6] \text{ 时 } |\varphi'(x)| \leq \frac{1}{3\sqrt[3]{(1+1.3^2)^2}} < 1$$

$$\text{且 } 1.3 < \sqrt[3]{1+1.3^2} \leq \varphi(x) \leq \sqrt[3]{1+1.6^2} < 1.6$$

$\therefore$ 迭代公式在区间 $[1.3, 1.6]$ 上收敛

13. 解: $\varphi(x) = 1 + 2Cx$

$\therefore$ 当 $x_{k+1} = \varphi(x_k)$ 具有局部收敛性时, $|\varphi'(x^*)| < 1$, 即 $-1 < 2Cx^* < 0$

原方程为 $x = x + C(x^2 - 3)$, 根为 $x = \pm\sqrt{3}$

则 C 的范围为 $(-\frac{1}{3}, 0) \cup (0, \frac{1}{3})$

22. 解: 假设 x^* 是 $f(x)$ 的单根, 即 $f(x^*) \neq 0$

$$\varphi(x) = x - \frac{[f(x)]^2}{f(x+f(x)) - f(x)}$$

$$\varphi'(x^*) = 1 - \frac{2f(x^*) \cdot f'(x^*)}{[f(x^* + f(x^*)) - f(x^*)]}, \quad = 1 \neq 0$$

$\therefore$ 构造出的迭代公式在 $f(x) = 0$ 的单根 x^* 附近为二阶收敛

习题五

1. 解: 雅可比迭代: $\begin{cases} x_1^{(k+1)} = \frac{1}{3}(2 - x_2^{(k)}) \\ x_2^{(k+1)} = \frac{1}{2}(1 - x_1^{(k)}) \end{cases}$, 计算结果如下

K	$x_1^{(k)}$	$x_2^{(k)}$	$ x_1^{(k)} - x_1^{(k-1)} $	$ x_2^{(k)} - x_2^{(k-1)} $	$< 0.0005?$
0	0	0	—	—	
1	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{1}{2}$	N
2	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{3}$	N
3	$\frac{11}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{12}$	N
4	$\frac{7}{12}$	$\frac{7}{36}$	$\frac{1}{36}$	$\frac{1}{18}$	N
5	0.60185	0.20833	0.01852	0.01389	N
6	0.59722	0.19908	0.00463	0.00925	N
7	0.60031	0.20139	0.00309	0.00231	N
8	0.59954	0.19985	0.00077	0.00154	N
9	0.60005	0.20023	0.00051	0.00038	N
10	0.59992	0.19998	0.00003	0.00025	Y

$$\therefore x_1^* \approx x_1^{(10)} \approx 0.600 \quad x_2^* \approx x_2^{(10)} \approx 0.200$$

高斯-赛德尔迭代: $\begin{cases} x_1^{(k+1)} = \frac{1}{3}(2 - x_2^{(k)}) \\ x_2^{(k+1)} = \frac{1}{6} - \frac{1}{2}x_1^{(k+1)} + \frac{1}{2} = \frac{1}{6}(1 + x_1^{(k+1)}) \end{cases}$

计算结果如下表.

k	$x_1^{(k)}$	$x_2^{(k)}$ $x_2^{(k)}$	$ x_1^{(k)} - x_1^{(k-1)} $	$ x_2^{(k)} - x_2^{(k-1)} $	$< 0.0005?$
0	0	0	—	—	
1	$\frac{2}{3}$	$\frac{1}{6}$ $\frac{1}{6}$	$\frac{2}{3}$	$\frac{1}{6}$	N
2	$\frac{11}{18}$	$\frac{7}{36}$	$\frac{1}{18}$	$\frac{1}{36}$	N
3	$\frac{65}{108}$	$\frac{47}{216}$	$\frac{1}{108}$	$\frac{1}{216}$	N
4	0.6003	0.1998	0.0015	0.0008	N
5	0.6000	0.2000	0.0003	0.0001	Y

$$x_1^* \approx x_1^{(5)} \approx 0.600 \quad x_2^* \approx x_2^{(5)} \approx 0.200$$

3. 证明: 由题可知, 系数矩阵为严格对角占优矩阵, 故方程组的雅可比迭代与高斯-塞德尔迭代均收敛

雅可比迭代:

$$x^{(k+1)} = D^{-1} (b - (L+U)x^{(k)})$$

$$D = \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 11 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

$$L = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 \\ 0 & 3 & -1 & 0 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

n	0	1	2
$x^{(n)}$	$[0, 0, 0, 0]^T$	$[0.6, 2.2727, -1.1, 1.875]^T$	$[0.94545, 2.06818, -1.00818, 1.0389]^T$

3	4
$[1.01211, 1.97154, -0.94061, 0.99142]^T$	$[1.00219, 1.94625, -0.94801, 0.99847]^T$

5

$$[1.00046, 1.99918, -0.99962, 0.99960]^T$$

由于 $x^{(5)}$ 与 $x^{(4)}$ 的差值小于 10^{-3} , 则 $x \approx (1.000, 2.000, -1.000, 1.000)^T$

高斯-塞德尔

$$x^{(k+1)} = (D+L)^{-1} (b - Ux^{(k)})$$

则:

$$x^{(0)} = [0, 0, 0, 0]$$

$$x^{(1)} = \begin{bmatrix} 0.6 \\ 2.32727 \\ -0.48727 \\ 0.87886 \end{bmatrix}$$

$$x^{(2)} = \begin{bmatrix} 0.94382 \\ 1.92731 \\ -1.02001 \\ 0.96203 \end{bmatrix}$$

$$x^{(3)} = \begin{bmatrix} 1.00117 \\ 1.99504 \\ -0.99716 \\ 0.99636 \end{bmatrix}$$

$$x^{(4)} = \begin{bmatrix} 1.00000 \\ 1.99940 \\ -0.99974 \\ 0.99962 \end{bmatrix}$$

由于 $x^{(4)}$ 和 $x^{(3)}$ 差值小于 10^{-3} , $x \approx (1.000, 2.000, -1.000, 1.000)^T$

5. 解: 1) 雅可比迭代:

$$\begin{cases} x_1^{(k+1)} = -2x_2^{(k)} - 1 \\ x_2^{(k+1)} = -3x_1^{(k)} + 2 \end{cases}, \|G\|_\infty = 3 > 1, \text{不收敛}$$

高斯-塞德尔亦不收敛

(2) 迭代矩阵为

$$G = \begin{bmatrix} -5 & 3 \\ \frac{5}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}$$

$\|G\|_\infty = 8 > 1$, 不收敛, 故无法用雅可比和高斯-塞德尔求解

7. 解: 利用高斯-塞德尔迭代:

$$\begin{cases} \tilde{x}_1^{(k+1)} = -\frac{3}{4}x_2^{(k)} + 4 \\ \tilde{x}_2^{(k+1)} = -\frac{3}{4}\tilde{x}_1^{(k+1)} + \frac{1}{4}x_3^{(k)} + 5 \\ \tilde{x}_3^{(k+1)} = \frac{1}{4}\tilde{x}_2^{(k+1)} - 3 \end{cases}$$

引入松弛因子, 得

$$\begin{cases} x_1^{(k+1)} = (1-\omega)x_1^{(k)} + \omega\tilde{x}_1^{(k+1)} = -\frac{1}{4}x_1^{(k)} + \frac{5}{4}\tilde{x}_1^{(k+1)} \\ x_2^{(k+1)} = -\frac{1}{4}x_1^{(k)} + \frac{5}{4}\tilde{x}_2^{(k+1)} \\ x_3^{(k+1)} = -\frac{1}{4}x_2^{(k)} + \frac{5}{4}\tilde{x}_3^{(k+1)} \end{cases}$$

计算结果见下表:

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$< \frac{1}{2} \times 10^{-4}$
0	0	0	0	N
1	5	2.5	-3.125	N
2	1.40625	2.65625	-2.14844	N
3	2.15820	3.03223	-2.28882	N
4	1.61173	3.15872	-2.19860	N
5	1.63577	3.24423	-2.19187	N
6	1.54959	3.28508	-2.17800	N
7	1.53284	3.30793	-2.17001	N
8	1.51561	3.31978	-2.16847	N
9	1.50880	3.32615	-2.16717	N
10	1.50453	3.32951	-2.16664	N
11	1.50245	3.33130	-2.16674	N
12	1.50129	3.33225	-2.16694	N
13	1.50069	3.33276	-2.16672	N
14	1.50037	3.33306	-2.16676	N
15	1.50016	3.33318	-2.16670	N
16	1.50010	3.33325	-2.16668	N
17	1.50005	3.33333	-2.16667	Y

迭代解 $x_1^* = x_1^{(17)} \approx 1.50005$, $x_2^* = x_2^{(17)} \approx 3.33333$, $x_3^* = x_3^{(17)} \approx -2.16667$

题六

$$1. (2) \text{解} \begin{cases} 3x_1 - x_2 + 2x_3 = -3 \quad ① \\ x_1 + x_2 + x_3 = -4 \quad ② \\ 2x_1 + x_2 - x_3 = -3 \quad ③ \end{cases}$$

$$\begin{aligned} ② - \frac{1}{3}①, \text{得} \\ ③ - \frac{2}{3}① \end{aligned}$$

$$\begin{cases} 3x_1 - x_2 + 2x_3 = -3 \quad ① \\ \frac{4}{3}x_1 + \frac{1}{3}x_3 = -\frac{13}{3} \quad ④ \\ \frac{5}{3}x_1 - \frac{7}{3}x_3 = -3 \quad ⑤ \end{cases}$$

$$④ - \frac{5}{4}⑤, \text{得} \begin{cases} 3x_1 - x_2 + 2x_3 = -3 \quad ① \\ \frac{3}{4}x_2 + \frac{1}{3}x_3 = -3 \quad ⑥ \\ x_3 = -1 \quad ⑦ \end{cases}$$

将⑦代入⑥, 得 $x_2 = -2$, 将⑦, ⑥代入①, 得 $x_1 = 1$

$$\therefore \begin{cases} x_1 = 1 \\ x_2 = -2 \\ x_3 = -1 \end{cases}$$

$$2. (1) \text{解} \left(\begin{array}{ccc|c} 1 & -1 & 1 & -4 \\ 5 & -4 & 3 & -12 \\ 2 & 1 & 1 & 11 \end{array} \right) \xrightarrow{r_1 \leftrightarrow r_2} \left(\begin{array}{ccc|c} 5 & -4 & 3 & -12 \\ 1 & -1 & 1 & -4 \\ 2 & 1 & 1 & 11 \end{array} \right) \xrightarrow{-\frac{1}{5}r_1 + r_2}$$

$$\left(\begin{array}{ccc|c} 5 & -4 & 3 & -12 \\ 0 & -\frac{1}{5} & \frac{2}{5} & -\frac{8}{5} \\ 2 & 1 & 1 & 11 \end{array} \right) \xrightarrow{5 \times r_2} \left(\begin{array}{ccc|c} 5 & -4 & 3 & -12 \\ 0 & -1 & 2 & -8 \\ 2 & 1 & 1 & 11 \end{array} \right) \xrightarrow{-\frac{2}{5}r_1 + r_3} \left(\begin{array}{ccc|c} 5 & -4 & 3 & -12 \\ 0 & -1 & 2 & -8 \\ 0 & \frac{13}{5} & -\frac{1}{5} & \frac{79}{5} \end{array} \right)$$

$$\xrightarrow{5 \times r_3} \left(\begin{array}{ccc|c} 5 & -4 & 3 & -12 \\ 0 & -1 & 2 & -8 \\ 0 & 13 & -1 & 79 \end{array} \right) \xrightarrow{r_2 \leftrightarrow r_3} \left(\begin{array}{ccc|c} 5 & -4 & 3 & -12 \\ 0 & 13 & -1 & 79 \\ 0 & -1 & 2 & -8 \end{array} \right) \xrightarrow{-\frac{1}{13}r_2 + r_3} \left(\begin{array}{ccc|c} 5 & -4 & 3 & -12 \\ 0 & 13 & -1 & 79 \\ 0 & 0 & \frac{25}{13} & -\frac{25}{13} \end{array} \right)$$

$$\frac{13}{25}r_3 \rightarrow \left(\begin{array}{ccc|c} 5 & -4 & 3 & -12 \\ 0 & 13 & -1 & 79 \\ 0 & 0 & 1 & -1 \end{array} \right)$$

$$\therefore x_3 = -1, x_2 = \frac{x_3 + 79}{13} = 6, x_1 = \frac{4x_2 - 3x_3 - 12}{5} = 3$$

3. 解: 由公式

$$\begin{cases} l_1 = b_1, \mu_1 = \frac{c_1}{l_1} \\ m_i = a_i \\ l_i = b_i - m_i \mu_{i-1} \\ \mu_i = c_i / l_i \end{cases}$$

得分解式 $A=LU$ 的元素为: $m_2 = a_2 = -1, l_1 = b_1 = 2, \mu_1 = c_1 / l_1 = -\frac{1}{2};$

$$m_3 = a_3 = -2, l_2 = b_2 - m_2 \mu_1 = \frac{5}{2}, \mu_2 = c_2 / l_2 = -\frac{4}{5};$$

$$m_4 = a_4 = -3, l_3 = b_3 - m_3 \mu_2 = \frac{12}{5}, \mu_3 = c_3 / l_3 = -5/4; l_4 = b_4 - m_4 \mu_3 = 5/4$$

追的过程为

$$y_1 = d_1 / l_1 = 3, y_2 = (d_2 - m_2 y_1) / l_2 = \frac{8}{5}, y_3 = (d_3 - m_3 y_2) / l_3 = \frac{1}{2}, y_4 = (d_4 - m_4 y_3) / l_4 = 2.$$

求的过程为

$$x_4 = y_4 = 2, x_3 = y_3 - m_3 x_4 = 3, x_2 = y_2 - m_2 x_3 = 4, x_1 = y_1 - m_1 x_2 = 5.$$

所以方程组的解为 $x = (5, 4, 3, 2)^T$

9. 解: 1) 由题意得 $A^{-1} = \begin{bmatrix} 9 & -36 & 30 \\ -36 & 144 & -180 \\ 30 & -180 & 180 \end{bmatrix}$

$$\therefore \text{cond}(A) = \|A\|_{\infty} \cdot \|A^{-1}\|_{\infty} = \frac{11}{6} \times 408 = 748$$

2) 由题意得

$$B^{-1} = \begin{bmatrix} 64 & 264 & -220 \\ -240 & 900 & 720 \\ 180 & -720 & 600 \end{bmatrix}$$

$$\text{cond}(B) = \|B\|_{\infty} \cdot \|B^{-1}\|_{\infty} = \frac{13}{12} \times 1860 = 2015$$